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Searching–Waiting Strategy: A Candidate for an Evolutionary Model of Depression?

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The model proposed here assumes that depressive disorders could reflect an extreme state of a current behavioral strategy. According to this model, a subject facing a problem of survival without apparent solution may choose between two behavioral patterns: searching for a solution or waiting for that solution to occur. This choice can be made after one or several estimations of the cost and benefit attached to each of these alternatives. Two of the main behavioral models of depression are interpreted according to this model: namely, infant response to maternal separation in monkeys and "behavioral despair" in rodents. Practical and theoretical consequences of this model are discussed.

Any living being works toward an end (von Bertalanffy, 1968) and therefore it uses strategies, i.e., goal-directed combinations of behavior patterns. According to neo-Darwinian evolution theory, behavior is bound by natural selection (Williams, 1966). The result is that the chosen strategies are outcomes of design operations which aim at the optimal compromises between the competing selective pressures characteristic of the environment (McFarland, 1976). Evolutionary interpretation of behavior thus leads to identification of ultimate causes, i.e., adaptive function of behavior, in animal and man (cf. McGuire & Essock-Vitale, 1981; McGuire, Essock-Vitale, & Polsky, 1981).

In this paper, we are concerned with the relationship between depressive disorders and adaptive functions. Depressive symptoms have been described both in human and in nonhuman species and, in spite of wide variations on account of species and context, they have been described as part of a common biphasic behavioral response where subjects may

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exhibit either activity ("protest," "agitation") or inactivity ("despair," "depression") (Katz, 1981; Mineka & Suomi, 1978; Stéru, 1981).

According to Engel (1962a, 1962b), faced with threat from the environment, the central nervous system is organized to mediate two opposite patterns of response to a mounting need. The first, broadly subsumed under the heading of flight–fight, involves activity, energy expenditure, and interactions with the environment to try to control sources of supply. The second, conservation–withdrawal, involves inactivity, energy conservation, raising of the stimulus barrier, and withdrawal from the environment. Each of these is seen as an inborn system, each with its own underlying mediating neural organization. The conservation–withdrawal system is a second biological defense organization which comes into play if and when the energy expenditures of the first reaction (flight–fight) threaten the organism with exhaustion before supplies are secured.

Kaufman and Rosenblum (1967) further elaborated this model and extended it to the protest and despair stages occurring in infant monkeys following separation from the mother. However, they continue to interpret flight–fight and conservation–withdrawal responses in terms of two separate and successive patterns. In contrast, Katz (1981) suggests that it should be possible to conceive these responses as a continuum of disrupted adaptation. Following Katz' suggestion, we propose here to consider the biphasic behavioral response as a result of a behavioral strategy. In this view, depressive behavior might be interpreted as part of a strategy we name the searching–waiting strategy.

In a first section, we shall describe the model. We do not imply that it is an actual model of depression, but its connections with depressive disorders will be suggested. In two subsequent sections, we shall use it to interpret the actually most successful animal models of depression, i.e., infant response to maternal separation in monkeys (Harlow & Suomi, 1974; Kaufman, 1977) and "behavioral despair" in rats and mice (Porsolt, 1981; Porsolt, Bertin, & Jalfre, 1977).

DESCRIPTION OF THE SEARCHING–WAITING STRATEGY MODEL

We assume that whenever a depressive symptomatology occurs, there is a common factor: *the subject faces a problem of survival which has no apparent solution.* By this assumption, we mean that the subject *estimates* on account of its information from the environment and its internal state, that it faces a problem without immediate solution and which jeopardizes its life. It must be emphasized that no hypothesis is made in our model upon the estimation mechanisms which may be conscious or not.

In such a situation, two behavioral patterns are possible for the subject:

1. *Searching pattern:* The subject *acts* in order to find a solution: it may use to this end any means at its disposal.

2. *Waiting pattern:* The subject *does not act* and waits for a solution to occur: it saves resources in order to wait as long as possible.

The subject may choose one strategy, i.e., one behavioral pattern, or switch alternatively from one to the other. A cost–benefit analysis (Mishan, 1971; Prest & Turvey, 1965) may be used to make statements about this choice. Searching and waiting patterns differ in their consequences: they have different costs and benefits, depending on the individual and environmental variables. The time–energy budget of the subject is limited, and therefore it has to behave in order to maximize benefit/cost ratio. The choices made depend on the relative costs and benefits estimated by the subject to be attached to each pattern.

Relative costs. Searching represents a higher level of activity than necessary for maintaining the organismic homeostasis. On the contrary, the subject only dispenses minimal energy in waiting. Searching may thus be assumed to be a more expensive pattern than waiting: $C_s > C_w$.

Relative benefits. The natural environment is rich in all kinds of events: changes are common, and there is an actual probability of success (i.e., solution occurrence) for the waiting subject. However, by its activity, the searching subject increases the probability of success, and it thus may be assumed that searching can produce a solution in a shorter time than waiting: $B_s > B_w$.

In summary, a searching pattern is expensive but generally provides a higher probability of success, while a waiting pattern is less expensive and provides a lower probability of success. The choice of a strategy is made after estimation of relative ratios: when $B_s/C_s > B_w/C_w$, then searching is chosen; when $B_s/C_s < B_w/C_w$, then waiting is chosen.

Individual and environmental variables vary in the course of time. Therefore, the subject must constantly update its estimation of cost and benefit. It may thus shift strategies, so that a repeated switch between searching and waiting patterns can be observed.

Finally, it must be emphasized that, for the convenience of this actual demonstration, searching and waiting patterns have been supposed to be two qualitatively different patterns. In fact, the difference is only quantitative, i.e., the subject may, for example, locomote more or less, or increase or decrease its body temperature. Searching–waiting strategy may be conceptualized as a continuum along which the subject's behavior varies between two poles, searching and waiting.

INTERPRETATION OF INFANT RESPONSE TO MATERNAL SEPARATION IN MONKEYS

Since Harlow's first studies upon mother–infant separation in macaques, numerous studies have been conducted in this field. Typically, the separated infant is reported to exhibit two successive stages: "protest" and "despair" (Dolhinow, 1980; Harlow & Suomi, 1974; Mineka & Suomi, 1978; Reite, Short, Seiler, & Pauley, 1981).

Protest stage. After separation, the infant gives distress calls, and its locomotor activity increases markedly. Following Kaufman (1977), we assume that the infant is searching: according to the searching–waiting strategy model, the infant searches for a solution, i.e., parental or alloparental care.

Despair stage. Within 24–36 h following the separation, social withdrawal and a marked decrease of activity occur. According to Kaufman (1977) "this is not depletion of energy or massive fatigue because in certain circumstances the infant moved rapidly and appropriately, e.g., when food was put into the pen or when an act of aggression was directed at it." Conservation is achieved through the marked inactivity, as well as by the posture (body almost rolled into a ball) which reduces the exposed body surface. Contrary to the hyperactivity and vocalization of the protest stage, the paucity and extreme retardation of movement decrease the risk of provoking an attack either from conspecifics or from predators.

While the adaptive interpretation of the protest stage does not seem to arouse any question, the conservation function attributed by Kaufman to the despair stage is criticized by Rosenblum and Plimpton (1981). Their arguments may be outlined under two headings: first, a depressed subject has almost no chance of survival in the natural environment; second, a unitary process of protest–despair cannot account for the observed variability of responses.

Chances of survival. A first argument of Rosenblum and Plimpton (1981) is that "in the natural environment of most nonhuman primates, the danger from predators, and the need to forage for food and water would make the chances of survival of a young infant in the absence of the mother and social group minimal." Therefore, depressive behavior should not be considered as adaptive.

If no change occurs, it is right that depression always leads to death. In fact, observations in the field have shown that mortality rate is very high in infants of which the mother has disappeared (dead or lost) but some survive (Altmann, 1980; Hamilton, Busse, & Smith, 1982; Hasegawa & Hiraiwa, 1980; Rhine, Norton, Roertgen, & Klein, 1980; Teleki, Hunt, & Pfifferling, 1976). Probability of survival is very low and therefore selective pressure is important. A difference of 1 or 2 days may be significant and this is why the separated infant must husband its resources to wait as long as possible for a solution. The solution may be either locating the mother or, in most cases for orphans, being adopted by a member of the troop (Hamilton et al., 1982, Hasegawa & Hiraiwa, 1980; Hrdy, 1976).

A second argument of Rosenblum and Plimpton (1981) is that, in separation studies carried out in the laboratory, infants sometimes die during the separation period (Dolhinow, 1980; Rosenblum, 1971). Actually, behavioral response to maternal separation is accompanied by physiological changes which consist on the whole of a decrease in metabolic activity

(Reite et al., 1981). It may be assumed that, facing an extreme situation, subjects present a borderline response. In this process, a few may present too intense a response and so they die: that could be considered as a pathological response.

Protest–despair as a unitary process. According to Rosenblum and Plimpton (1981), there is a discontinuity between the initial and subsequent patterns of responses to separation, because changes in experimental variables may cause protest and despair to vary in opposite directions. These authors support their view with two facts: first, repeated 4-day peer separation is reported to result in diminution of initial protest and in increase in some measures of subsequent despair (Mineka & Suomi, 1978). According to the searching–waiting strategy model, the more a given problem seems without a solution to the subject, the more this subject must wait; i.e., it must be less "agitated" and more "depressed." The second fact is that the removal of an infant to a new environment in conjunction with separation induces relatively heightened levels of protest, but fewer indications of subsequent despair. On the contrary, when the infant is left in its home pen, and the mother is removed, depressive state appears appreciably earlier than usual (Chappel & Meier, 1975; Hinde & Davies, 1972; Hinde & Mc Ginnis, 1977; Rosenblum & Plimpton, 1981). Such data can be explained in terms of the searching–waiting model: in the first case, the infant is separated and displaced in an unknown milieu, and the more effective behavior is then to relocate the home range and the mother, i.e., being "especially agitated"; in the second case, the infant remains in its home range, and the more effective behavior is to wait for the mother to come back, i.e., being "early depressed."

Further evidence of protest–despair as a unitary process may be provided. As pointed out by earlier authors (Lewis, McKinney, Young, & Kraemer, 1976; Mineka & Suomi, 1978), although all species which have been investigated show at least a biphasic reaction to separation, there are wide variations in the form, severity, and duration of reaction to separation. Several factors acting on this variability have been identified: species, ontogenetic experience, and duration, number, and context of separations. Action of these factors is influenced by the very nature of the process: according to the searching–waiting model, the subject may vary along a continuum of which protest and despair are only two typical facets. Indeed, while some infants, for example, markedly diminish their locomotor activity in the despair stage, others alternate periods of hyperactivity with periods of inactive sitting (Spencer-Booth & Hinde, 1971). In addition, this process may account for the apparent discrepancies frequently found according to the criteria used in assessment of protest and despair stages (Dolhinow, 1980; Lewis et al., 1976; Reite et al., 1981; Rosenblum and Plimpton, 1981): while some physiological indices of depression (heart

rate, body temperature), for example, show maximal decrease early in separation, some behavioral indices (decreases in play or social grooming) tend to be maximal later in the separation.

Monkey depression bears a striking resemblance to some causal and behavioral aspects of human depression. It is not possible to develop here this assumption, demonstrated elsewhere (Stéru, 1981). The causal point of view in humans is the most difficult to assess, but it may be remembered that the breaking of an attachment bond (Bowlby, 1977) and inadequacy when facing a change (Haynal, 1976) are identified as important factors in triggering reactive depression in man. Monkey depression can also be alleviated by antidepressant treatments, like antidepressant drugs (Hrdina, van Kulmiz, & Stretch, 1979; Suomi, Collins, Harlow, & Ruppenthal, 1978) and electroconvulsive therapy (McKinney, 1977). In the following section we investigate another animal analog where causal analogs and behavioral symptoms may seem very far from human depression but where antidepressant treatments are also effective.

INTERPRETATION OF "BEHAVIORAL DESPAIR" TEST IN RATS AND MICE

The test called "behavioral despair" has been described by Porsolt (1977) and Porsolt et al. (1981) in rats and mice. When these rodents are forced to swim in a restricted space (e.g., in a narrow receptacle containing water), they first show vigorous activity: swimming movements around the circumference of the receptacle and climbing movements against the inner surface. After a few minutes, immobility periods appear: subjects remain floating passively in the water, making only the necessary movements for keeping the head out of water. Activity periods may still alternate with immobility periods, but the duration of the latter increases in time.

Porsolt et al. suggested that the immobility observed in water might reflect a "state of despair" and showed that immobility was selectively reduced by a variety of treatments which are therapeutically effective in depression: antidepressant drugs, electroconvulsive shock, and REM sleep deprivation. These authors also showed that antidepressants reduced immobility even at doses which reduced spontaneous locomotion, for sedative drugs, and at doses which did not modify spontaneous locomotion, for the stimulant drugs.

Following Porsolt et al., we assume that "behavioral despair" tests in rats and mice may be considered as an animal model sharing some common points with human depression (that is, mainly psychomotor inhibition in a solutionless aversive environment). In addition, we propose to interpret it according to the searching–waiting strategy model: in activity periods, the subject tries to escape, i.e., it is searching for an exit; in immobility periods, the subject floats, passively, i.e., it is waiting

for a change to occur. It must be emphasized that such a behavior may be adaptive under natural conditions. As an example, let us consider a wild subject that has just fallen in a river. It may choose between two patterns, either swimming in order to try to reach the bank, or floating in order to let itself be carried by the current. The optimal solution depends on two kinds of factors: environmental factors (strength and direction of the current, temperature of the water, distance from the bank, etc.) and individual factors (physical strength, internal state, knowledge about the environment, etc.). Most factors may change during the episode, and, on account of the available information, the subject's behavior may constantly alternate from activity to inactivity.

Regarding Porsolt's test, it may be objected that laboratory conditions differ from ecological ones. In the test, the subject is never given any opportunity to swim before the test session; moreover, it has no chance to escape from the receptacle. However, one must keep in mind that the subject cannot embrace all the factors of the situation: it chooses a behavioral strategy only on account of its *estimation* of the situation. The experimental situation is a novel experience for the subject, and it seems most likely that the absence of solution to the situation is not obvious from the beginning of the test. This can account for the vigorous swimming periods that initiate the test.

DISCUSSION

What can be common between a healthy animal (facing a test situation) and a depressed patient? Several arguments stand for the opinion that there is no real model for human depression. Following McKinney (1977), such a model should satisfy four criteria: (1) comparable symptomatology, (2) comparable etiology, (3) comparable neurophysiological background, and (4) concordant effect of drugs. No model has yet simultaneously satisfied these criteria. In addition, the effect of antidepressants in the tests as well as in the human disease is still difficult to understand. In the animal models, the drugs show an effect (reducing immobility) within half an hour. But it is well known that the depressed patient has to wait for 2 weeks, on average, before depression is alleviated by the drugs. If there is something in common between man and animal, it is not easy to envisage the parallel between pathology and test situation.

However, the drug action upon models and diseases leads to the conclusion that there is a biological bridge from man to animal. It is not possible to evoke here all the literature on the biology of depression that supports this assumption. It is sufficient to remember that every model of depression proposed in animals, including learned helplessness (Seligman, 1975), motor incapacitation (Weiss, Glazer, & Pohorecky, 1976), and even separation in monkeys (see for review Porsolt, 1981; Mineka & Suomi, 1978) show evidence of a catecholaminergic deficit that is

concordant with the biochemical hypothesis for human depression (Akiskal & McKinney, 1973; Schildkraut, 1965).

This paper has sought to interpret two of the main models of depression in terms of searching–waiting strategy. This has been done for mother–infant separation, for "behavioral despair," and the same could be attempted for learned helplessness experiments (Katz, 1981; Maier & Seligman, 1976; Seligman, 1975). It becomes possible to assume that the searching–waiting strategy hypothesis shares not only a behavioral relationship but also a biochemical similitude with depressive disorder. We consider here depressive disorder as a whole, without debating between the different clinical types, thus considering a common depressive feature as a "final common pathway" (Akiskal & McKinney, 1973).

It must be emphasized that searching–waiting strategy is not equivalent to depressive disorders but may be a model of adaptive behavior, of which depressive disorders are a pathologic state. Searching–waiting strategy represents a continuum of the behavioral patterns which may be chosen by a subject facing a problem of survival without apparent solution. Choices are made on account of the estimated energetic costs and the expected benefits (probability of success) related to each behavioral pattern. The estimation is repeated in time, resulting in an alternation of the two types of strategy.

In man, feelings of sadness or even depressive states may be induced by various changes in life situations (mourning, birth of first child, retirement, moving home); such moods are generally temporary, and they may reflect the waiting pattern. Depression, in contrast, refers to sustained states of deep dejection accompanied by lowered self-esteem, suicidal preoccupations, and serious somatic dysfunctions. It is tempting to conceptualize depression as a specific pathology of the natural balance between searching and waiting patterns. According to McGuire et al. (1981), normal individuals may be characterized by variability in situations where individuals suffering from psychiatric disorders often repeat the same behaviors, i.e., have a deficit in changing strategies.

A reading of the existing literature on depression from an ethological point of view has led to a model based on an evolutionary interpretation of behavior. There has already been an empirical validation (Stéru, Chermat, Thierry, & Simon, in press). Multidisciplinary testing is necessary to assess the heuristic value of this theoretical framework.

REFERENCES

Akiskal, H. S., & McKinney, W. T. (1973). Depressive disorders: Toward a unified hypothesis. *Science*, **182,** 20–29.

Altmann, J. (1980). *Baboon mothers and infants*. Cambridge: Harvard Univ. Press.

Bertalanffy, L. von (1968). *General systems theory*. New York: Braziller.

Bowlby, J. (1977). The making and breaking of attachment bond. I. Aetiology and psychopathology in the light of attachmennt theory. *British Journal of Psychiatry*, **130,** 201–210.

Chappell, P. F., & Meier, G. W. (1975). Modification of response to separation in the infant rhesus macaque through manipulation of the environment. *Biological Psychiatry,* **10,** 643–653.

Dolhinow, P. (1980). An experimental study of mother loss in the Indian langur monkey *(Presbytis entellus). Folia primatologica,* **33,** 77–128.

Engel, G. L. (1962a). Anxiety and depression withdrawal: The primary affects of unpleasure. *International Journal of Psycho-analysis,* **52,** 183–196.

Engel, G. L. (1962b). *Psychological development in health and disease.* Philadelphia: Saunders.

Hamilton, W. J., Busse, C., & Smith, K. S. (1982). Adoption of infant orphan chacma baboons. *Animal Behaviour,* **30,** 29–34.

Harlow, H. H., & Suomi, S. J. (1974). Induced depression in monkeys. *Behavioral Biology,* **12,** 273–296.

Hasegawa, T., & Hiraiwa, M. (1980). Social interactions of orphans observed in a free-ranging troop of Japanese monkeys. *Folia primatologica,* **33,** 129–158.

Haynal, A. (1976). *Le sens du désespoir.* Paris: PUF.

Hinde, R. A., & Davies, L. (1972). Removing infant rhesus from mother for 13 days compared with removing mother from infant. *Journal of Child Psychology and Psychiatry,* **13,** 227–237.

Hinde, R. A., & Mc Ginnis, L. (1977). Some factors influencing the effects of temporary mother–infant separation: Some experiments with rhesus monkeys. *Psychological Medicine,* **7,** 197–212.

Hrdina, P. D., Kulmiz, P. van, & Stretch, R. (1979). Pharmacological modification of experimental depression in infant macaques. *Psychopharmacology,* **64,** 89–93.

Hrdy, S. B. (1976). The care and exploitation of non-human primate infants by conspecifics other than the mother. *Advances in the Study of Behavior,* **6,** 101–168.

Katz, R. J. (1981). Animal models and human depressive disorders. *Neuroscience and Biobehavioral Reviews,* **5,** 231–246.

Kaufman, I. C. (1977). Developmental considerations of anxiety and depression: Psychobiological studies in monkeys. *In* T. Shapiro (Ed.), *Psychoanalysis and contemporary science* (pp. 317–363). New York: International Univ. Press.

Kaufman, I. C., & Rosenblum, L. A. (1967). The reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. *Psychosomatic Medicine,* **29,** 648–675.

Lewis, J. K., McKinney, W. T., Young, L. D., & Kraemer, G. W. (1976). Mother–infant separation in rhesus monkeys as a model of human depression: A reconsideration. *Archives of General Psychiatry,* **33,** 699–705.

Maier, S. F., & Seligman, M. E. P. (1976). Learned helplessness: Theory and evidence. *Journal of Experimental Psychology (General),* **105,** 3–46.

McFarland, D. J. (1976). Form and function in the temporal organisation of behaviour. *In* P. P. G. Bateson & R. A. Hinde (Eds.), *Growing points in ethology* (pp. 55–93). Cambridge: Cambridge Univ. Press.

McGuire, M. T., & Essock-Vitale, S. M. (1981). Psychiatric disorders in the context of evolutionary biology: A functional classification of behavior. *Journal of Nervous and Mental Disease,* **169,** 672–686.

McGuire, M. T., Essock-Vitale, S. M., & Polsky, R. H. (1981). Psychiatric disorders in the context of evolutionary biology: An ethological model of behavioral changes associated with psychiatric disorders. *Journal of Nervous and Mental Disease,* **169,** 687–704.

McKinney, W. T. (1977). Biobehavioral models of depression in monkeys. *In* I. Hanin & E. Usdin (Eds.), *Animal models in psychiatry and neurology.* Oxford: Pergamon Press.

Mineka, S., & Suomi, S. J. (1978). Social separation in monkeys. *Psychological Bulletin.* **85,** 1376–1400.

Mishan, E. J. (1971). *Cost-benefit analysis.* New York: Praeger.

Porsolt, R. D. (1981). Behavioral despair. *In* S. J. Enna, J. B. Malick, & E. Richelson (Eds.), *Antidepressants: Neurochemical, behavioral and clinical perspectives* (pp. 121–139). New York: Raven Press.

Porsolt, R. D. Bertin, A., & Jalfre, M. (1977). Behavioural despair in mice: A primary screening test for antidepressants. *Archives Internationales de Pharmacodynamie et de Thérapie,* **229,** 327–336.

Prest, A. R., & Turvey, R. (1965). Cost-benefit analysis: A survey. *Economical Journal,* **75,** 683–735.

Reite, M., Short, R., Seiler, C., & Pauley, J. D. (1981). Attachment, loss, and depression. *Journal of Child Psychology and Psychiatry,* **22,** 141–169.

Rhine, R. J., Norton, G. W., Roertgen, W. J., & Klein, H. D. (1980). The brief survival of free-ranging baboon infants (*Papio cynocephalus*) after separation from their mothers. *International Journal of Primatology,* **1,** 407–409.

Rosenblum, L. A. (1971). Infant attachment in monkeys. *In* H. R. Schaffer (Ed.), *The origins of human social relations* (pp. 85–108). New York: Academic Press.

Rosenblum, L. A., & Plimpton, E. H. (1981). The infant's effort to cope with separation. *In* M. Lewis & L. A. Rosenblum (Eds.), *The uncommon child* (pp. 225–257). New York: Plenum.

Schildkraut, J. (1965). Catecholamine hypothesis of affective disorders. *American Journal of Psychiatry,* **122,** 509–522.

Seligman, M. E. P. (1975). *Helplessness: On depression, development and death.* San Francisco: Freeman.

Spencer-Booth, Y., & Hinde, R. A. (1971). Effects of 6 days separation from mother on 18- to 32-week-old rhesus monkeys. *Animal Behaviour,* **19,** 174–191.

Stéru, L. (1981). *Dépression et comportement de résignation.* Thèse de doctorat en médecine, Université Pierre et Marie Curie, Paris.

Stéru, L., Chermat, R., Thierry, B., & Simon, P. (in press). Tail suspension test (T. S. T.): A new method for screening antidepressants in mice. *Psychopharmacology.*

Suomi, S. J., Collins, M. L., Harlow, H. F., & Ruppenthal, G. C. (1976). Effects of maternal and peer separations on young monkeys. *Journal of Child Psychology and Psychiatry,* **17,** 101–112.

Teleki, G., Hunt, E. E., & Pfifferling, J. H. (1976). Demographic observations (1963–1973) on the chimpanzees of Gombe National Park, Tanzania. *Journal of Human Evolution,* **5,** 559–598.

Weiss, J. M., Glazer, H. I., & Pohorecky, L. A. (1976). Coping behavior and neurochemical changes: An alternative explanation for the original "Learned Helplessness" experiments. *In* G. Serban & A. Kling (Eds.), *Animal models in psychiatry and neurology.* Oxford: Pergamon.

Williams, G. C. (1966). *Adaptation and natural selection: A critique of some current evolutionary thought.* Princeton, NJ: Princeton Univ. Press.